Technology Stack

Date	15 March 2026
Team ID	XXXXXX
Project Name	LegalEase
Maximum Marks	2 Marks

Define Your Technology Stack:

The LegalEase solution uses a Python-based web architecture with Streamlit for the frontend, FastAPI for the backend, Google Gemini for AI-powered legal document generation, and document formatting libraries for TXT, DOCX and PDF export. The supplied project does not specify a database or a version-control/CI-CD technology.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit	Provides the user-facing web interface for entering document type, parties, terms and effective date. It also provides document preview, editing and download controls.
2	Backend / Server-Side	Python + FastAPI + Uvicorn	FastAPI handles API requests and exposes the POST /generate endpoint. Python supports the application logic and AI/document-processing modules, while Uvicorn runs the FastAPI application.
3	Database / Data Storage	No database specified	The supplied LegalEase project does not define MySQL, MongoDB, PostgreSQL or another database. The documented workflow processes user inputs and generated documents without a specified persistent database.
4	Cloud / Hosting / Deployment	Local deployment; future options: Render, Railway, Deta, Streamlit Community Cloud, Fly.io or VPS	The documented deployment milestone focuses on local deployment using FastAPI and Streamlit. The project also lists these services as possible future deployment options.
5	Version Control & CI/CD	Not specified	No Git, GitHub, GitLab, Jenkins or other version-control/CI-CD technology is specified in the supplied project document.

6	Third-Party APIs / Other Tools	Google Gemini API / SDK; python-docx; FPDF; Pillow; requests; python-dotenv	Gemini generates the legal content. python-docx creates formatted Word documents, FPDF creates branded PDFs, Pillow supports image/logo handling, requests supports HTTP requests, and python-dotenv manages environment configuration.

Technology Selection Summary: The documented architecture consists of Streamlit → FastAPI → Gemini AI, with dedicated formatting utilities for DOCX, PDF and HTML preview. The project specifies Python 3.10+, a .env file for the Gemini API key, and dependencies installed through pip/requirements.txt.